Final Technical Report

Autonomous Constellation Manager for Space Debris Risk Mitigation

Team MindMatrix

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1 Problem and Objectives

The system is designed to continuously monitor a constellation, predict close approaches with debris, and trigger safe maneuvers under fuel and operations constraints. The practical objectives are:

1. maintain real-time situational awareness,
2. detect conjunctions at scale,
3. minimize collision risk while preserving mission continuity,
4. provide transparent operator controls and telemetry.

2 Numerical Methods

2.1 State Dynamics

Each orbiting object is modeled with ECI state vector

$$\mathbf{x} = [x, y, z, v_x, v_y, v_z]^T.$$

The propagation model combines two-body gravity with J2 perturbation:

$$\dot{\mathbf{r}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = -\mu \frac{\mathbf{r}}{\|\mathbf{r}\|^3} + \mathbf{a}_{J2}.$$

The J2 acceleration is

$$\mathbf{a}_{J2} = \frac{3J_2\mu R_E^2}{2\|\mathbf{r}\|^5} \begin{bmatrix} x\left(5\frac{z^2}{\|\mathbf{r}\|^2} - 1\right) \\ y\left(5\frac{z^2}{\|\mathbf{r}\|^2} - 1\right) \\ z\left(5\frac{z^2}{\|\mathbf{r}\|^2} - 3\right) \end{bmatrix}.$$

2.2 Time Integration

A fixed-step RK4 method is used:

$$\begin{aligned} \mathbf{k}_1 &= f(\mathbf{x}_n), \\ \mathbf{k}_2 &= f\left(\mathbf{x}_n + \frac{\Delta t}{2}\mathbf{k}_1\right), \\ \mathbf{k}_3 &= f\left(\mathbf{x}_n + \frac{\Delta t}{2}\mathbf{k}_2\right), \\ \mathbf{k}_4 &= f\left(\mathbf{x}_n + \Delta t \mathbf{k}_3\right), \\ \mathbf{x}_{n+1} &= \mathbf{x}_n + \frac{\Delta t}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \end{aligned}$$

Operationally, long user-requested steps are chunked to keep API responsiveness stable.

2.3 Initialization

Initial states are produced from TLE catalogs via SGP4 at epoch t_0:

$$(\mathbf{r}_0, \mathbf{v}_0) = \mathrm{SGP4}(\mathrm{TLE}, t_0).$$

3 Spatial Optimization and Risk Algorithms

3.1 KD-Tree Candidate Screening

Brute-force pair search is avoided by KD-tree broad-phase filtering. For satellites set $\mathcal{S}$ and debris set $\mathcal{D}$ at time step t_k:

$$\mathcal{C}_k = \{(i,j) \in \mathcal{S} \times \mathcal{D} : \|\mathbf{r}_i(t_k) - \mathbf{r}_j(t_k)\| \leq r_{\mathrm{screen}}\}.$$

This reduces screening from all-pairs complexity toward practical near-neighbor search.

3.2 Collision Probability Approximation

For short encounters, collision probability uses a Chan-style approximation:

$$P_c \approx \frac{R^2}{2\sigma^2} \exp\left(-\frac{d^2}{2\sigma^2}\right),$$

where d is miss distance, R is hard-body radius, and σ is position uncertainty scale.

3.3 Maneuver Optimization Under Constraints

Fuel update follows Tsiolkovsky:

$$\Delta m = m_0 \left(1 - \exp\left(-\frac{\Delta v}{I_{sp} g_0}\right)\right).$$

A practical planning score is

$$J = w_1 \Delta d_{\min} - w_2 \|\Delta \mathbf{v}\| - w_3 \Delta m - w_4 \tau_{\mathrm{outage}},$$

subject to

$$\begin{aligned} \|\Delta \mathbf{v}\| &\leq \Delta v_{\max}, \\ t_{k+1}^{\mathrm{burn}} - t_k^{\mathrm{burn}} &\geq t_{\mathrm{cooldown}}, \\ m_{\mathrm{fuel}} - \Delta m &\geq 0. \end{aligned}$$

4 System Architecture

4.1 Layered Design

4.2 Operational Workflow

5 Implementation and Submission Notes

- API contract endpoints are implemented and tested: telemetry, maneuver scheduling, and simulation step.
- Frontend controls were hardened with queued stepping and guarded auto-run to prevent overlapping requests.
- Dockerized runtime is configured for port 8000 and deployment-ready launch.

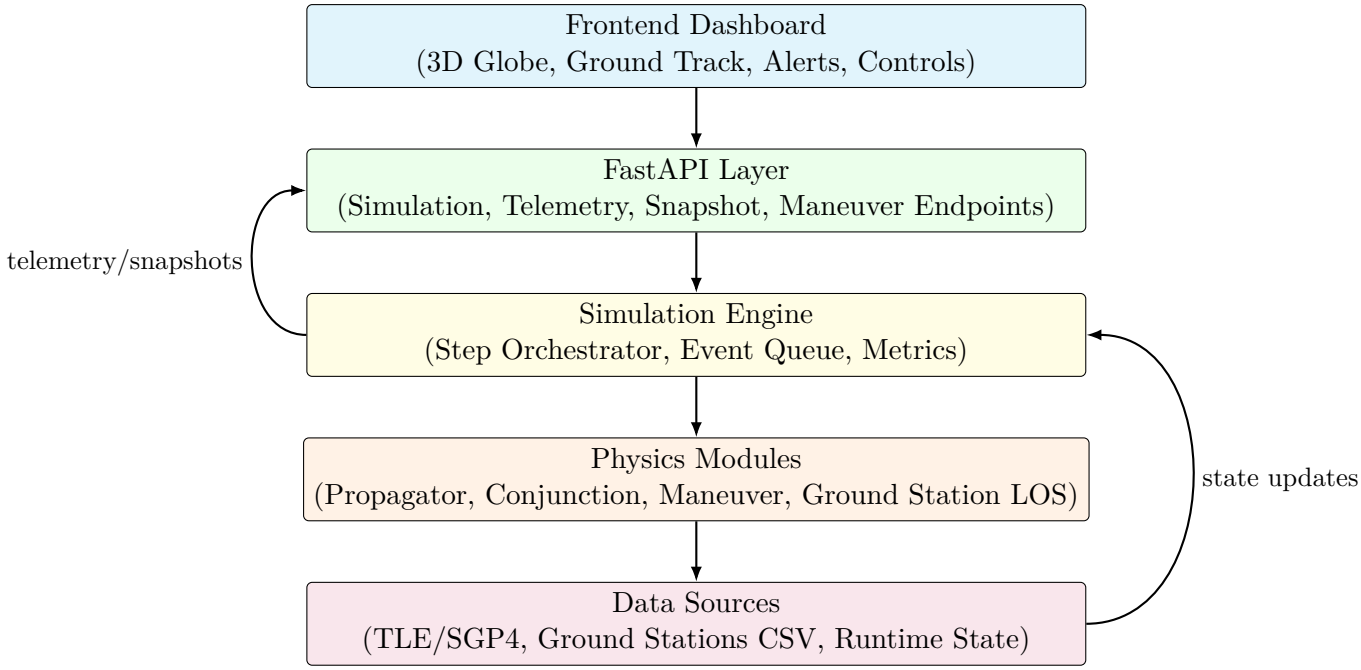

Figure 1: End-to-end architecture implemented by Team MindMatrix.

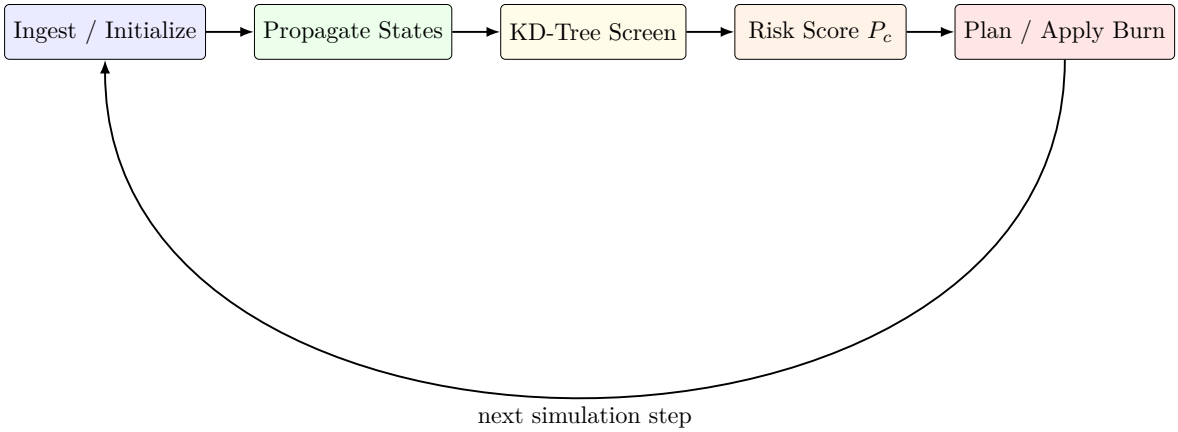

Figure 2: Autonomous simulation-control loop.

6 Conclusion

Team MindMatrix's final system couples reliable numerical propagation with scalable spatial risk screening and constrained control planning in a clean service architecture. The resulting platform is both technically rigorous and operationally usable for live mission-style demonstrations.

Key Symbols

Symbol	Meaning
μ	Earth gravitational parameter
R_E	Earth equatorial radius
J_2	second zonal harmonic coefficient
P_c	collision probability
d	miss distance
I_{sp}	specific impulse
g_0	standard gravity